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Dictyostelium RacH Regulates Endocytic Vesicular Trafficking and is Required for Localization of Vacuolin

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***Dictyostelium* RacH localizes predominantly to membranes of the nuclear envelope, endoplasmic recticulum and Golgi apparatus. To investigate the role of this protein, we generated knockout and overexpressor strains. RacH-deficient cells displayed 50% reduced fluid-phase uptake and a moderate exocytosis defect, but phagocytosis was unaffected. Detailed examination of the endocytic pathway revealed defective acidification of early endosomes and reduced secretion of acid phosphatase in the presence of sucrose. The distribution of the post-lysosomal marker vacuolin was altered, with a high proportion of cells showing a diffuse vesicular pattern in contrast to the wild-type strain, where few intensely stained vacuoles predominate. Cytokinesis, cell motility, chemotaxis and development appeared largely unaffected. In a cell-free system, RacH stimulates actin polymerization, suggesting that this protein is involved in actin-based trafficking of vesicular compartments. We also investigated the determinants of subcellular localization of RacH by expression of green-fluorescent-protein-tagged chimeras in which the C-terminus of RacH and the plasma-membrane-targeted RacG were exchanged, the insert region was deleted or the net positive charge of the hypervariable region was increased. We show that several regions of the molecule, not only the hypervariable region, determine targeting of RacH. Overexpression of mistargeted RacH mutants did not recapitulate the phenotypes of a strain overexpressing nonmutated RacH, indicating that the function of this protein is in great part related to its subcellular localization.**

Key words: actin, chemotaxis, cytokinesis, *Dictyostelium*, endocytosis, vacuolin, hypervariable region, isoprenylation, Rho GTPase

Rho guanosine triphosphatases (GTPases) constitute a major branch of the Ras superfamily of small GTPases. They function as molecular switches that cycle between an active GTP-bound form and an inactive GDP-bound form. Cycling is tightly regulated by a large number of GTP exchange factors (GEFs) and GTPase activating proteins (GAPs) and by binding to GTP dissociation inhibitors (GDIs). On activation, the Rho GTPases interact with downstream target proteins to induce cellular responses. Although primarily associated with cytoskeletal remodeling, Rho GTPases have been shown to be also involved in the regulation of a long list of cellular processes that include endocytosis, vesicle trafficking, morphogenesis, cytokinesis, transcriptional activation, cell cycle progression and cellular proliferation (1). In fact, some members of the Rho family have emerged as important regulators of several stages of vesicle trafficking (2,3). The roles of Rac1, Ccd42 and RhoA in the initial phase of endocytosis in mammalian cells are well established, particularly during phagocytosis and clathrin-independent internalization, although there is also evidence for a regulatory role in clathrin-mediated endocytosis. However, the actions of Rho GTPases are not restricted to the plasma membrane. Diverse studies point to a function of specific Rho GTPases in endocytic trafficking pathways in intracellular compartments. For example, RhoD regulates the motility of early endosomes (4), whereas RhoB is involved in transport to lysosomes (5). Cdc42 also functions at the Golgi apparatus, where it is recruited through interaction with the COPI vesicle coat protein coatomer and triggers Arp2/3-dependent actin assembly (6). Finally, several studies have put forward the relevance of Rho GTPases in exocytosis. For example, the Cdc42-related TC10 is required for translocation of Glut4 from storage vesicles to the plasma membrane (7).

In the model organism *Dictyostelium discoideum*, the Rho family comprises 18 members (8–10). In addition to Rho GTPases, other components of Rho-regulated signal transduction pathways known in mammalian cells are also present in this organism, like RhoGDI, the Arp2/3 complex, PI 3-kinases, PAK, WASP, Scar/WAVE, formins and a large number of GEFs and GAPs (9,11). This makes *Dictyostelium* an attractive model to investigate the signaling events that elicit remodeling of the cytoskeleton. While only few of the *Dictyostelium* Rho GTPases have been studied to some extent, the studies published to date give an idea of the repertoire of cellular processes regulated by Rho GTPases

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in this organism (see refs. 10 and 11 for reviews). RacB and the three Rac1 proteins Rac1a, Rac1b and Rac1c have been most extensively studied. They are involved in chemotaxis, cell motility, endocytosis, cytokinesis and development. RacC plays roles in actin cytoskeleton organization, pinocytosis, phagocytosis and cytokinesis. RacE appears to be essential for cytokinesis and actin polymerization but is not involved in phagocytosis. RacF1 localizes to early phagosomes, macropinosomes and transient cell-to-cell contacts, but inactivation of the *racF1* gene does not impair endocytosis and other actin-dependent processes, probably due to the presence of a closely related RacF2. In most cases, the exact signaling pathways regulated by these proteins remain largely unknown.

Like Ras, Rho GTPases are synthesized as cytosolic proteins but have the capacity to associate with membranes by virtue of a series of posttranslational modifications of a C-terminal CAAX sequence motif (where C is cysteine, A represents aliphatic amino acids and X is any amino acid). This motif signals for covalent attachment of an isoprenoid lipid unit to the cysteine residue. Two distinct soluble CAAX prenyltransferases have been described in eukaryotic cells: farnesyl transferase (FTase) and geranylgeranyl transferase (GGTase). The FTase catalyzes the addition of a C15 farnesyl unit to CAAX sequences where X is serine, methionine, cysteine, alanine or glutamine. The GGTase catalyzes the addition of a C20 geranylgeranyl unit to CAAX sequences where X is leucine or isoleucine. The prenyl-CAAX motif is substrate for a specific protease that cleaves the AAX tripeptide. The modified prenyl cysteine is then recognized by a specific carboxyl methyl transferase that methylesterifies the α carboxyl group. These two enzymes are membrane proteins localized in the endoplasmic recticulum (ER) (12). The CAAX motif alone targets proteins to the ER and Golgi apparatus (13). For transport from the ER to the plasma membrane, a second signal within the hypervariable region upstream of the CAAX motif of Ras proteins has been postulated, either palmitoylation (N-ras and H-ras) or a polybasic amino acid stretch (K-ras4B) (14). In the case of Rho GTPases, targeting is not only regulated by similar determinants in the hypervariable region but also by binding to RhoGDI and by additional factors that remain to be elucidated (15). More recently, it was shown that the type of lipid modification of the CAAX motif determines the necessity of postprenylation processing: Ras, usually farnesylated, requires further processing but not Rho, usually geranylgeranylated (16).

Posttranslational modification by isoprenoid lipids is not only essential for targeting, but it also appears to be critical for GTPase function. This was initially investigated in Ras proteins, where mutation of the CAAX motif or treatment with FTase inhibitors abolishes plasma membrane association and causes a complete loss of oncogenic transforming activity (14). The functional relevance of protein prenylation for Rho GTPases has been studied in RhoA and RhoB: nonprenylated versions of activated RhoA or RhoB displayed a loss of transforming activity (17,18), and the type of prenylation in part determines the function of RhoB: farnesylated RhoB is a growth promoting protein, whereas geranylgeranylated RhoB is an apoptosis-inducing protein (19,20).

In this study, we have undertaken the functional analysis of *Dictyostelium* RacH. Although not a true Rac protein, RacH is closer to Rac and Cdc42 (up to 74% similarity) than to other subfamilies like Rho, Rif or Rnd (51–63% similarity) (8,10). To investigate the role of this protein, we have generated knockout cell lines and cell lines that overexpress wild type (WT), constitutively active or dominant negative RacH fused to green fluorescent protein (GFP). RacH is targeted to intracellular membranes, predominantly of the nuclear envelope, ER and Golgi apparatus. RacH appears to be primarily involved in endosomal trafficking and has no impact on cell motility or chemotaxis. The function of RacH is in great part related to its subcellular localization, as revealed by the functional analysis of mistargeted RacH mutants. In a cell-free system, RacH stimulates actin polymerization, suggesting that this protein might be involved in actin-based trafficking of vesicular compartments.

Results

Knockout and overexpression of RacH

To investigate the function of RacH *in vivo*, we have generated a knockout strain (RacH-KO) and stably transformed cell lines that overexpress the WT and two mutated variants of this protein fused to a GFP tag. Mutation of Met-13 to Val (V13) renders the Rac protein constitutively active, whereas mutation of Thr-18 to Asn (N18) results in a dominant negative protein. The *racH* gene was deleted by homologous recombination (Figure S1A), and the recombination event was verified by Southern blot (Figure S1B) and Western blot (Figure 1A). A polyclonal antiserum generated against RacH recognized a protein of approximately 25 kDa in total cell homogenates of AX2 that was absent in the knockout strain. Endogenous RacH is sometimes visible as a double band, probably representing unprocessed and posttranslationally modified protein. In cells overexpressing a GFP fussion of RacH-WT, the antiserum recognized an additional protein of approximately 50 kDa that corresponds to the predicted size of GFP-RacH (Figure 1A). The identity of this band was further confirmed using a monoclonal antibody (mAb) against GFP (Figure 1B). By comparison of the blots in Figure 1A,–B, we estimated the levels of GFP-RacH-WT, V13 and N18 to be more than 10-fold higher than the amount of endogenous RacH.

Subcellular localization of RacH

The subcellular localization of RacH was studied by confocal microscopy of vegetative cells overexpressing GFP-RacH. Observations of live cells revealed a diffuse staining all over the cell. However, after fixation, much of the RacH, probably the cytosolic fraction, was washed off, allowing the observation of the GFP fusion protein associated to

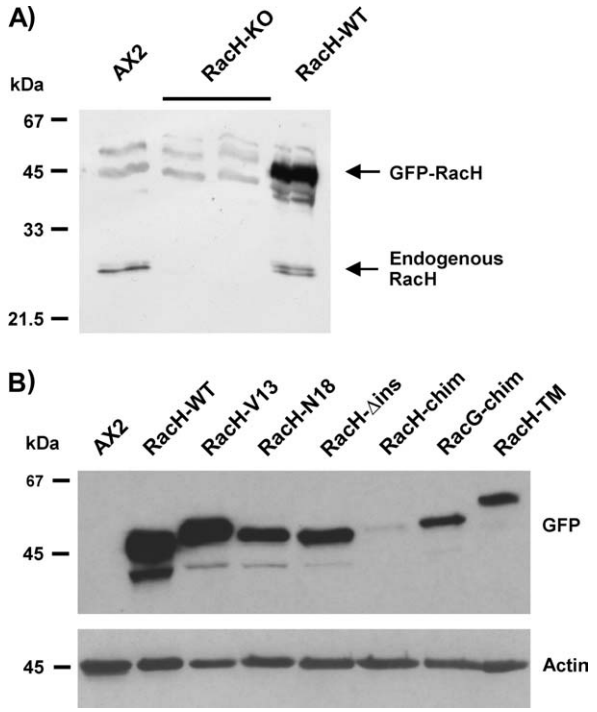

Figure 1: Expression levels of endogenous and GFP-tagged RacH. Total cell homogenates of 4×10^5 cells were resolved in polyacrylamide gels and blotted onto nitrocellulose membranes. A) Western blot analysis of AX2, RacH-KO (two independent clones) and GFP-tagged RacH-WT overexpressor. The blot was incubated with a polyclonal antiserum specific for RacH (1:2000 dilution). B) Western blot analysis of AX2 and strains overexpressing GFP fusions of WT, constitutively active (V13), dominant negative (N18), insert-deleted (Δins), and mitochondrial-membrane-targeted (TM) forms of RacH as well as chimeric constructs with RacG (see Figure 11). The blot was incubated with an anti-GFP antibody and reprobed with mAb Act 1-7 specific for actin as a control for loading.

internal membrane compartments (Figure 2). An association with the plasma membrane was observed in many cases, but it was difficult to resolve from the fluorescence signal of internal membranes close to the cell cortex (see also Figure 11C). To identify the membranous compartment(s) populated by RacH, we stained GFP-RacH cells with antibodies against various marker proteins: the nuclear envelope and ER marker interaptin, the Golgi marker comitin, and the ER marker protein disulfide isomerase (PDI). VatA, a subunit of the vacuolar proton pump adenosine triphosphatase (ATPase), was used as a marker for the contractile vacuole, where 90% of the ATPase is located, and the endo/lysosomal system, where the rest of the ATPase resides (21). Vacuolin was used as a marker for a postlysosomal compartment, and an antibody against a common mannose-6-sulfate-containing carbohydrate epitope was used as a lysosomal marker. GFP-RacH clearly colocalized with interaptin and comitin and to a great extent also with PDI. Colocalization with VatA, vacuolin and lysosomes was apparent in some instances but was not pronounced.

Morphology and actin distribution in RacH mutants

To examine the morphology and patterns of actin distribution of RacH mutants, we generated maximum projection images from confocal sections through cells fixed and stained with an actin-specific antibody (Figure 3). This procedure renders images with a pseudo-three-dimensional aspect. In AX2 cells, actin displays a characteristic smooth cortical staining, with enrichment at crown-like membrane protrusions and short filopods. RacH-KO cells were heterogeneous in size, with a high proportion of cells of small size, but there were no noticeable alterations in their pattern of actin distribution. When grown in suspension, RacH-WT cells were of irregular sizes (see also Figure S4), and many, especially the bigger ones, did not have well-developed crowns. RacH-V13 and N18 had a similar morphology as RacH-WT when grown in suspension (not shown). However, when grown on a substrate, these three mutants were indistinguishable from AX2.

Growth and endocytosis of RacH mutants

The pattern of subcellular localization of RacH and the fact that Rho GTPases participate in the regulation of endocytic and exocytic pathways prompted us to examine in detail vesicle trafficking processes in RacH mutants. We examined the ability of RacH mutant cells to internalize fluorescently labeled yeast particles (Figure 4A) or the fluid-phase marker tetramethylrhodamine isothiocyanate-dextran (TRITC-dextran) (Figure 4B). RacH-KO, V13 and N18 cells showed normal rates of particle uptake. By contrast, RacH-WT showed a severe defect, only reaching 30% of the AX2 rate.

RacH-KO cells and cells overexpressing RacH-V13 and RacH-N18 were able to internalize the fluid-phase marker TRITC-dextran to a lesser extent (approximately 50%) than the control AX2 cells, but in cells overexpressing RacH-WT, pinocytosis was almost abolished. The rates of pinocytosis roughly correlated with the growth rates of these mutants in suspension. RacH-WT cells had a severe growth defect: they were able to grow only to a density slightly more than 2×10^6 cells/mL, with a very prolonged doubling time (36.5 h). RacH-KO, V13 and N18 cells were slightly impaired: they grew slowly (10 to 12 h versus 8.5 h for AX2) and reached densities below that of AX2 cells (around 8×10^6 versus 10×10^6 cells/mL) (Figure S2). In the overexpressor mutants, a cytokinesis defect (see below) is also likely contributing to impaired growth in suspension. Finally, we quantified exocytosis rates of TRITC-dextran and found that RacH-KO cells showed a slower rate of exocytosis. RacH-V13 and N18 cells were able to release the marker at the same rate as AX2 cells, but RacH-WT cells were slightly impaired (Figure 4C).

Acidification of early endosomes is impaired in RacH-KO cells

As RacH seems to play a role in the regulation of endocytosis, we further analyzed the endosomal and vacuolar system of RacH-KO cells. First, we monitored the change of endosomal pH during early phases of

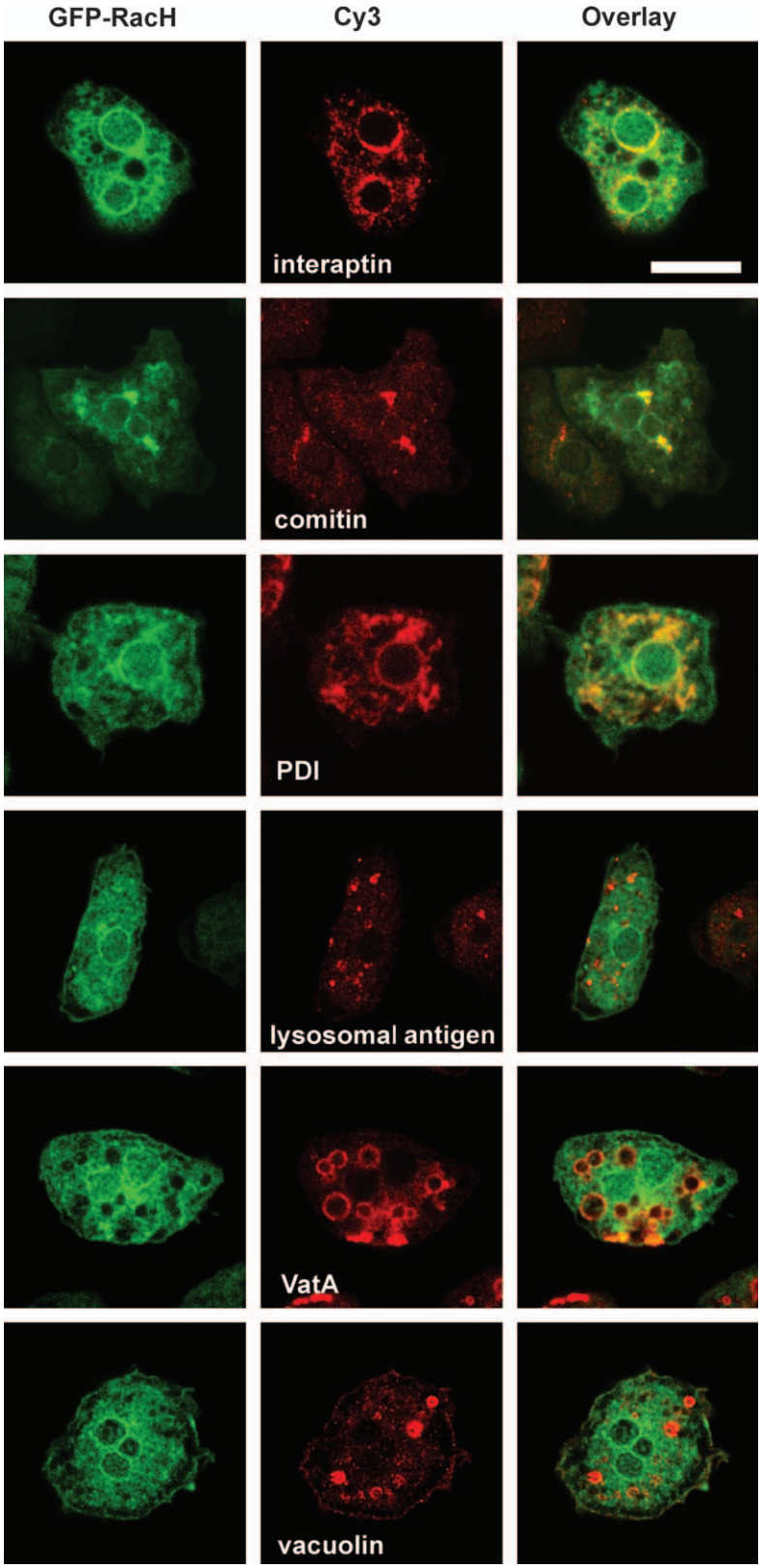

Figure 2: Subcellular distribution of RacH. RacH associates with intracellular membrane compartments, particularly with the nuclear envelope, the Golgi apparatus and the ER. Cells expressing GFP-RacH-WT were fixed in cold methanol and incubated with mAbs that recognize specific membrane compartments, followed by incubation with Cy3-labeled anti-mouse immunoglobulin G. The markers used were the nuclear envelope and ER marker interaptin (mAb 260-60-10), the Golgi marker comitin (mAb 190-340-2), the ER marker PDI (mAb 221-135-1), a lysosomal marker (mAb 221-342-5), a marker for the vacuolar H^+-ATPase (vatA, mAb 221-35-2) present at the contractile vacuole system and to a lesser extent at endosomes, and a marker for a postlysosomal compartment (vacuolin, mAb 221-1-1). GFP is visualized directly. Images are confocal sections. Bar, 10 μm.

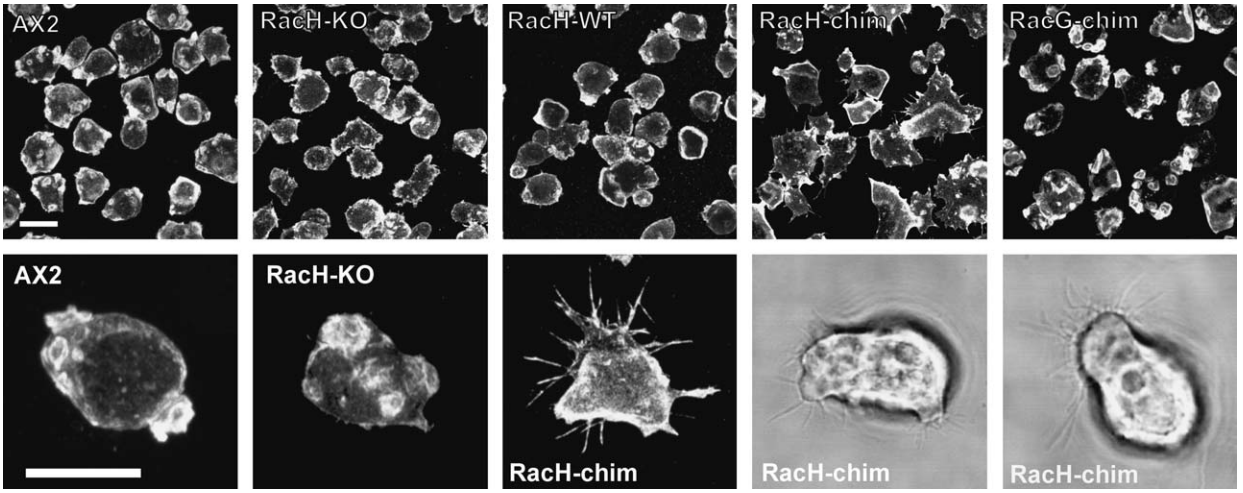

Figure 3: Cell morphology and F-actin organization of RacH mutants. Cells were allowed to sit on coverslips, fixed with picric acid/paraformaldehyde and stained with actin-specific mAb Act 1-7, followed by Cy3-labeled anti-mouse immunoglobulin G. In RacH-chim cells, long actin-rich filopods are abundant. The lower panels are closer views of a single AX2, RacH-KO and RacH-chim cell along with phase contrast images of living RacH-chim cells. Immunofluorescence pictures are the maximum projection of 20 confocal sections 400 nm apart. Scale bars represent 10 μm.

pinocytosis and phagocytosis following a 10-min pulse of fluorescein isothiocyanate-dextran (FITC-dextran) (Figure 5A) or FITC-labeled bacteria (Figure 5B). In AX2 cells, endosomal acidification was rapid and reverted during the next 30 min. By contrast, in RacH-KO cells, endosomes became on average less acidic. As endosomes are acidified by the V(H^+)-ATPase, we compared the levels of VatA, a component of the peripheral catalytic domain of the vacuolar proton pump, in RacH-KO cells (Figure 5C). The levels of VatA did not differ between RacH-KO and AX2 cells.

Additionally, we visualized pH changes during endosomal traffic *in vivo*. We exposed RacH-KO cells to medium containing pH-sensitive FITC-dextran and pH-independent TRITC-dextran. After 1 h, when cells have reached an equilibrium of endocytosis and exocytosis, images of both markers were collected (Figure 6A) and the distribution of acidic and neutral vesicles was calculated. We did not notice significant alterations: RacH-KO cells displayed 82.7% acidic and 17.3% neutral endosomes, and AX2 cells displayed 84.8% acidic and 15.2% neutral endosomes. Furthermore, we investigated the ability of endosomes to concentrate their contents by measuring the intensity of TRITC-dextran in vesicles compared to its concentration in the medium. The marker concentration in acidic and neutral endosomes of AX2 and RacH-KO cells did not differ significantly (Figure 6B).

Secretion of lysosomal enzymes

Like other lower eukaryotes, *Dictyostelium* cells partly secrete lysosomal enzymes, likely due to incomplete retrieval from the postlysosmal compartment (22). Measuring the activity of lysosomal enzymes provides a means to quantify the relative amounts of exocytosis versus recycling of lysosomal enzymes. Some lysosomal enzymes, like acid phosphatase, enter endosomes early in the pathway to profit from the acidic luminal pH value and may be efficiently recycled and reused. Other enzymes, like α-mannosidase, are delivered to endosomes later, when their pH has returned to neutral, and tend to be secreted at higher rates and thus accumulate in the medium. Secretion of lysosomal enzymes can be stimulated by nonmetabolizable sugars, like sucrose, that induce endosomal swelling, thus counteracting membrane retrieval processes (23). We compared the secretion of acid phosphatase (Figure 7A) and α-mannosidase (Figure 7B) in RacH-KO and AX2 cells. In the absence of sucrose, lysosomal secretion of both enzymes was comparable in RacH-KO and AX2 cells. Sucrose provoked an increase in lysosomal enzyme secretion rates, especially acid phosphatase. In the case of α-mannosidase, no major differences were found between RacH-KO and AX2 cells; by contrast, RacH-KO cells showed almost no increase of secreted phosphatase activity.

Subcellular distribution of vacuolin is affected in RacH-deficient cells

The pattern of subcellular localization of RacH and the trafficking defects observed in RacH-KO cells led us to investigate whether ablation of RacH would lead to alterations of the pattern of distribution of any of the compartments involved in transit of endocytic vesicles. In fixed cells, we studied the distribution of comitin, PDI, VatA, a lysosomal antigen and vacuolin. Except for vacuolin, none of these markers was noticeably altered in RacH-KO cells (Figure S3). By contrast, the vacuolin distribution of RacH-KO cells was remarkably distinct from the one of AX2 cells (Figure 8A). While 50% of AX2 cells exhibited one to three large vacuoles,

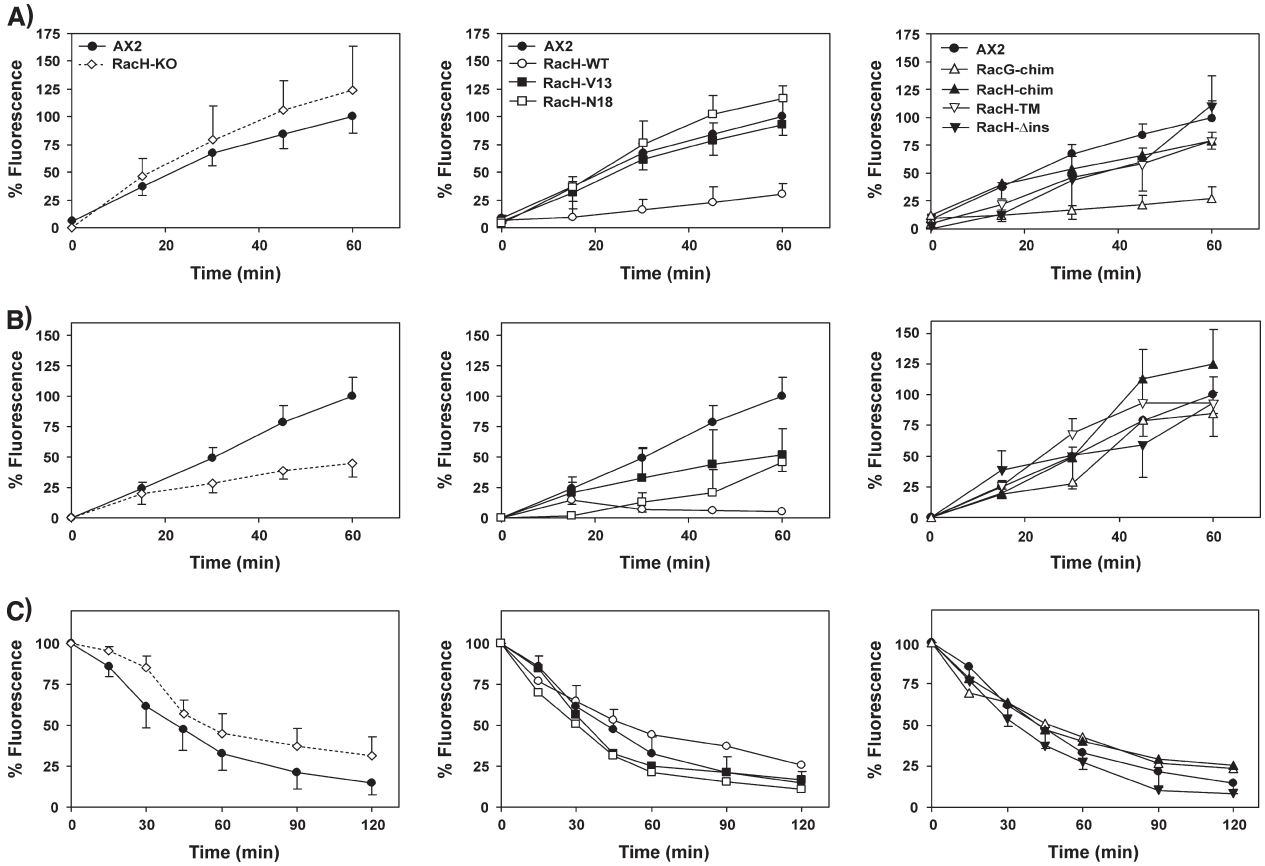

Figure 4: Phagocytosis, fluid-phase uptake and exocytosis of RacH mutants. A) Phagocytosis of TRITC-labeled yeasts. Cells were resuspended at 2×10^6 cells/mL in fresh axenic medium and challenged with fivefold excess fluorescent yeast cells. Fluorescence from internalized yeasts was measured at the designated time points. B) Fluid-phase endocytosis of TRITC-dextran. Cells were resuspended in fresh axenic medium at 5×10^6 cells/mL in the presence of 2 mg/mL TRITC-dextran. Fluorescence from the internalized marker was measured at selected time points. C) Fluid-phase exocytosis of TRITC-dextran. Cells were pulsed with TRITC-dextran for 3 h, washed, and resuspended in fresh axenic medium. Fluorescence from the marker remaining in the cell was measured. Data are presented as relative fluorescence, AX2 being considered 100%. All values are the average of at least three independent experiments. For the sake of clarity, error bars (standard deviation) are depicted in one direction, and in C, only error bars for AX2 and RacH-KO are shown.

only 22% of RacH-KO cells showed this morphology. In 66% of RacH-KO cells, but only in 23% of AX2 cells, vacuolin staining is dominated by a pattern of small vacuoles dispersed all over the cell (Figure 8B). This suggests that RacH is also involved in sorting of vacuolin to postlysosomal vesicles. Examination of the vacuolin distribution in overexpressor mutants supported this observation: in cells overexpressing RacH-WT and RacH-N18, the proportion of cells containing large vacuoles was severely reduced, and most cells had numerous middle-sized vacuoles. In RacH-V13 cells, the proportion of cells with small, intensely stained vacuoles was unusually large, and in many cases, vacuolin accumulated strongly at the plasma membrane (Figure 8A,B).

RacH directly induces actin polymerization

We investigated the ability of RacH to induce actin polymerization using a cell-free system (24). Under these conditions, addition of guanosine triphosphate gamma S (GTPγS) to lysates of RacH-V13 cells induced a higher actin polymerization (about 150%) when compared to addition of GTPγS to lysates of AX2 cells (about 120%). When the cell lysate was incubated with a RacH-specific polyclonal antiserum prior to addition of GTPγS, the response was similar to that of an AX2 lysate. This demonstrates the specificity of the response observed in RacH-V13 lysates (Figure 9A). The ability of RacH to induce actin polymerization was confirmed in a rhodamine actin-based visual assay. A trace amount of labeled actin was added to a cell lysate to follow the behavior of endogenous actin. Using a lysate of RacH-V13 cells, we observed that addition of GTPγS stimulated actin polymerization around membrane fragments containing GFP-RacH (Figure 9B).

Chemotaxis and cell motility

Aggregation competent cells move actively toward a cyclic AMP (cAMP) gradient. Stimulation with cAMP elicits fast and highly transient changes in the F-actin content that correlate with changes in cell behavior (25). Rho-regulated

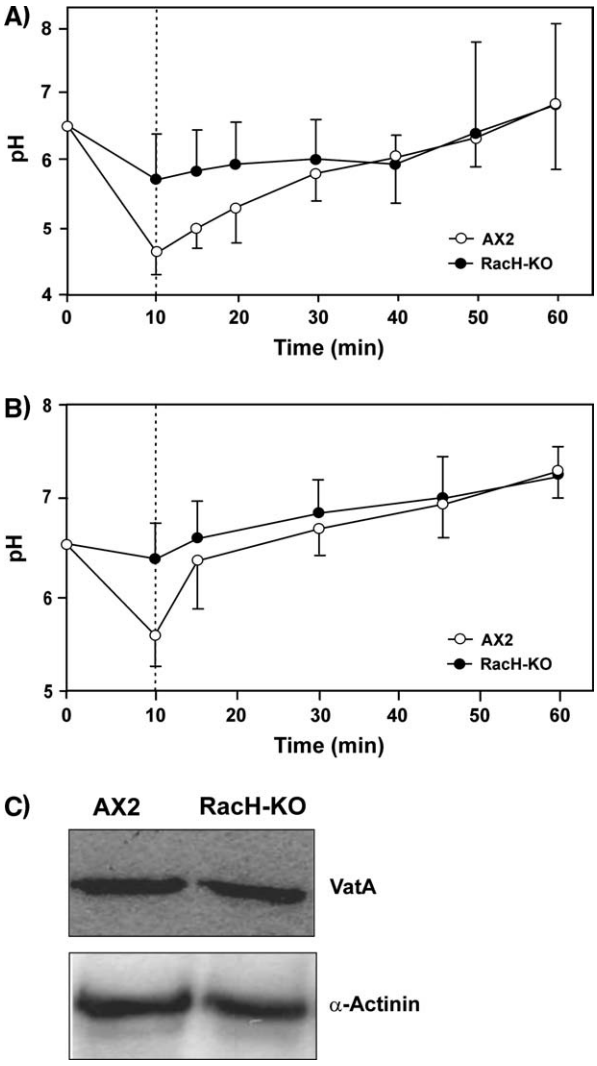

Figure 5: Change of endosomal pH during transit of a fluid-phase or a solid-phase marker. Cells were pulsed with FITC-dextran (2 mg/mL) A) or FITC-labeled *Escherichia coli* B/r (500:1 ratio) B) for 10 min, then washed and resuspended in fresh nutrient medium. Samples were collected at the indicated time points, and dual excitation ratio was used to calculate the endosomal pH. In the RacH-deficient cells, endosomes turn significantly less acidic than in the WT. C) VatA levels are comparable in AX2 and RacH-KO cells. Total cell homogenates of 4×10^5 cells were resolved in 12% polyacrylamide gels and blotted onto nitrocellulose membranes. The blot was probed with mAb 221-35-2 for detection of VatA and mAb 47-62-17 for labeling of α-actinin as loading control.

signaling pathways have been shown to be involved in this process (11,26). It was therefore of interest to investigate to what extent ablation or overexpression of RacH alters actin polymerization and the motile behavior of aggregation competent cells. In AX2 cells, stimulation with cAMP resulted in a rapid and transient 1.9-fold increase in the amount of F-actin, followed immediately by a second much lower peak that lasted until approximately 60 seconds. A similar response was observed in RacH-KO and RacH-WT cells (Figure 10A) as well as in RacH-V13 and RacH-N18 cells (not shown), although the second peak was abolished.

In a micropipette-based chemotaxis assay, AX2 and all other mutant strains exhibited a similar rate of locomotion (4–6 μm/min) in the absence of cAMP (Table 1). Parameters like persistence, directionality and directional change were indicative of random movement with frequent turns. In the presence of cAMP, AX2 and RacH-V13 behaved similarly, whereas RacH-KO, WT and N18 displayed a moderately lower speed (8.0–9.6 versus 11.8 μm/min of AX2). In general, RacH-KO cells performed moderately worse than AX2 cells in the presence of cAMP. All these strains became polarized, formed streams and migrated toward the tip of the micropipette (Figure 10B).

Cytokinesis and development of RacH mutants

Rho-mediated signaling pathways have been shown to participate in the regulation of cytokinesis in diverse eukaryotic organisms (27). We were therefore interested to investigate whether overexpression of RacH provoked any alteration in this process. We stained cells grown under shaking conditions or on a solid substrate with the DNA-binding dye 4′,6-diamidino-2-phenylindole (DAPI) (Figure S4). AX2 cells are usually mononucleate or binucleate under any of these conditions. RacH-KO cells were indistinguishable from AX2 cells. By contrast, in cultures of RacH-WT cells grown under shaking conditions, 30% of the cells had three or more nuclei. This defect was more severe in RacH-V13 and N18 cells, where almost 60% of the cells showed three or more nuclei [Figure S4A (arrows),B]. Interestingly, the distribution of nuclei in RacH-overexpressing strains grown on coverslips for 2 days did not differ from that in AX2 (not shown), indicating that the cytokinesis defect of these strains is conditional.

When starved on phosphate-buffered agar, RacH-KO cells did not differ significantly from AX2 cells in the timing and morphology of the developmental process (Figure S5). Development studies in the overexpressor strains were not attempted: whereas after 6 h of starvation, levels of the GFP fusions were substantial, they became undetectable after 12 h, paralleling the downregulation of the actin-15 promoter during development (28).

Determinants of RacH targeting

We sought to investigate the molecular determinants of the pattern of subcellular localization exhibited by RacH. In RacH, the last residue of the CAAX motif is glutamine instead of methionine. A CAAQ motif is a substrate for farnesylation, whereas a CAAM motif can be farnesylated or geranylgeranylated (29). However, a RacH mutant in which the last glutamine was exchanged by methionine (Q200M) displayed the same subcellular localization as the WT RacH (not shown). We then focused our attention on the hypervariable region. We analyzed the C-terminal

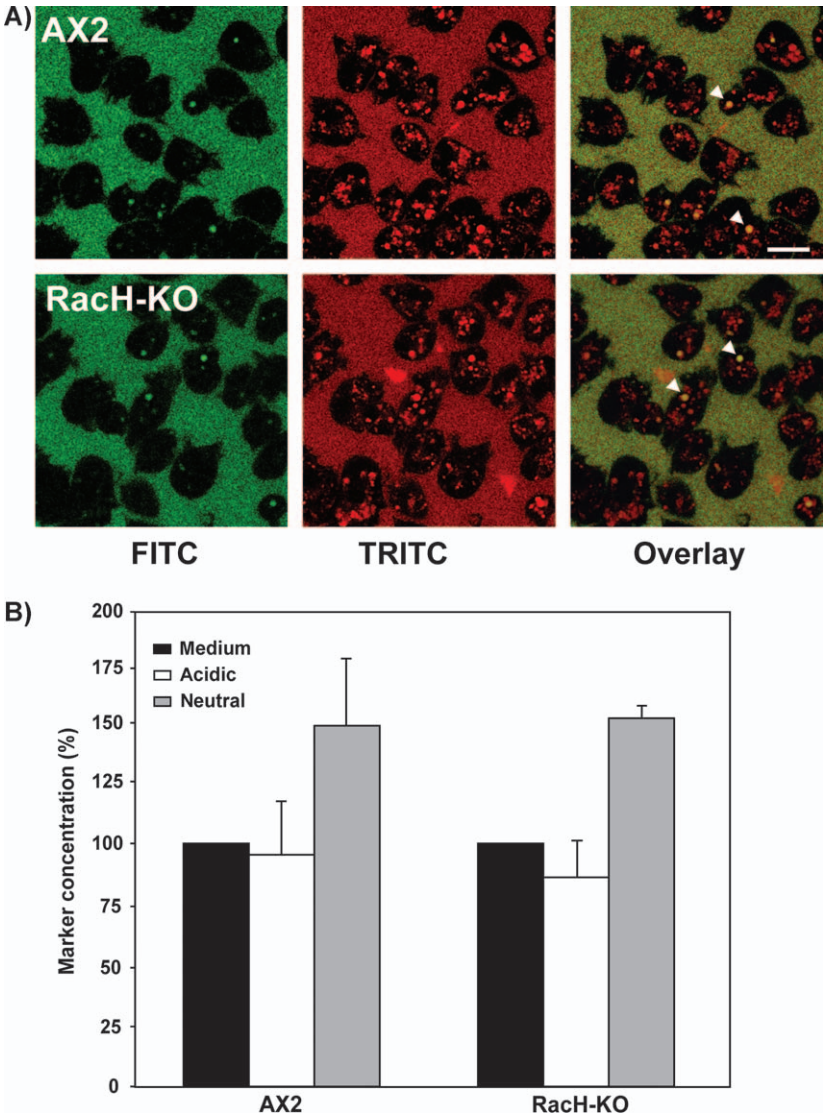

Figure 6: Visualization of pH changes during endosomal traffic *in vivo*. Wild-type and RacH-KO cells were allowed to sit on a glass surface and incubated with nutrient medium containing the pH-sensitive FITC-dextran and the pH-independent TRITC-dextran. A) Single confocal sections showing distribution of FITC-dextran (green) and TRITC-dextran (red). Superimposition of the channels yields a yellow color for neutral vacuoles (examples are marked with arrowheads), while acidic vesicles appear in red. Note that vesicular staining can be of higher intensity than extracellular medium, indicating concentration of marker during passage through the endocytic pathway. Scale bar, 10 μm. B) Marker concentration determined from acidic and neutral vesicles of AX2 and RacH-KO cells. Fluorescence intensity of TRITC-dextran in the vesicles was quantitated relative to the intensity in the medium from digital images similar to the ones shown in A) for more than 200 vesicles each. Values are average ± standard deviation.

sequence of RacH in comparison with Rho GTPases in *Dictyostelium* that are targeted predominantly to the plasma membrane, like RacF1 and RacG (Figure 11A). In RacF1 and RacG, numerous lysine residues confer a net charge of +6 and +8, respectively, to the hypervariable region. By contrast, in RacH, interspersed acidic residues lower the net charge to +2. To test the hypothesis that these charge differences are responsible for the differential localization of RacH and RacG, we made chimeric constructs in which the C-terminal regions of RacH and RacG were interchanged (Figure 11B). In confocal sections of fixed preparations, we noticed that RacH-chim was enriched at the plasma membrane, resembling RacG. RacG-chim distributed homogeneously throughout the cell including the nucleus (Figure 11C). To rule out that the strong plasma membrane association of RacG is due to additional lipid modification of the cysteine residue 183 (Figure 11A, arrow), we exchanged this residue by serine. This RacG C183S mutant behaved like the unmodified RacG, indicating that net positive charge is sufficient for plasma membrane targeting of this molecule. The charge effect was further investigated in alanine-exchange constructs. Here, aspartic acid residues of the hypervariable region of RacH were exchanged by alanine residues in order to increase the net positive charge of this region (Figure 11A, asterisks). Interestingly, both RacH-4Ala and RacG-chim-4Ala constructs displayed increased, but not predominant, accumulation at the plasma membrane (Figure 11C). A similar behavior was observed in a RacH-2Ala mutant (not shown). This indicates that net charge of the hypervariable region is important but is not the only determinant for proper targeting of RacH, suggesting the presence of an additional targeting signal within the C-terminal region of RacH.

We next investigated the role of the insert region. This region is characteristic of Rho proteins and required for activation of some effectors but not others. It is usually

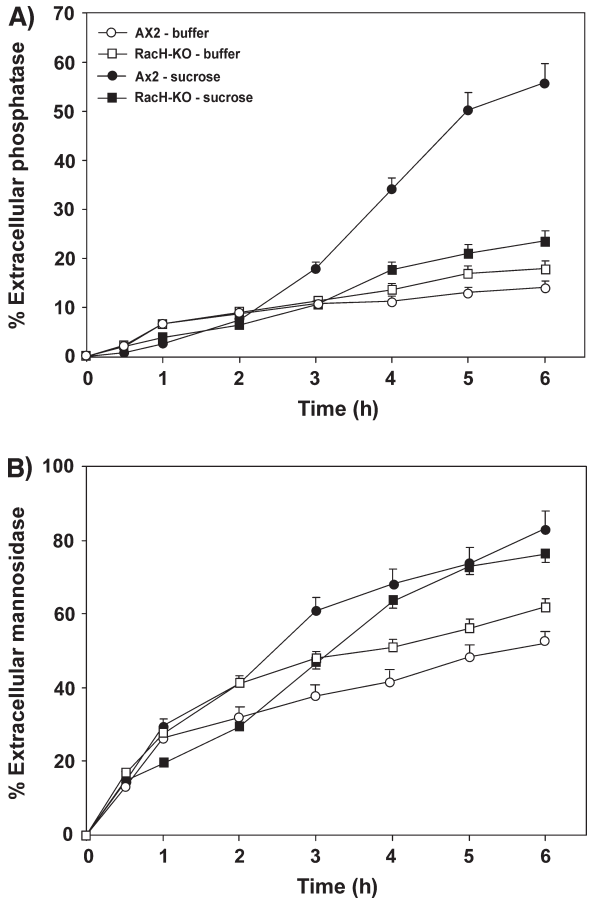

Figure 7: Secretion of lysosomal enzymes. Cells were harvested by centrifugation and resuspended at a concentration of 1×10^7 cells/mL in Soerensen phosphate buffer, with or without addition of sucrose (100 mM). The acid phosphatase A) and α-mannosidase B) activities were measured both in the supernatant and in the cell lysates at the indicated time points and results expressed as percentage of extracellular enzyme activity. Values are the average ± standard deviation of two independent experiments. In comparison to AX2, the RacH-KO mutant does not respond with higher acid phosphatase secretion in the presence of sucrose.

about 13 residues long, but in RacH, it only comprises six residues. Remarkably, insert-deleted RacH (RacHΔins) displayed a homogeneous pattern of localization, with enrichment in the cell nucleus, indicating that the insert region is essential for correct subcellular localization of RacH (Figure 11C).

The pattern of membrane association of these mutants was further studied by differential centrifugation of lysates, followed by Western blot analysis of the resultant cytosolic and particulate fractions (Figure 11B). In the case of RacH, approximately 60% of the protein was cytosolic and 40% membrane associated. RacG was predominantly membrane associated, 70% being present in the particulate fraction. These proportions became reversed in the chimeric mutants. The alanine-exchange mutants displayed an increased proportion of membrane-associated protein, particularly the RacH-4Ala mutant, whereas the RacHΔins mutant appeared equally distributed between the cytosolic and the membrane fractions.

Proper targeting of RacH is required for function

Because exchange of the C-terminal region leads to mislocalization of the protein, we sought to investigate how overexpression of a mislocalized protein affects diverse Rho-regulated processes like morphology, endocytosis, cytokinesis, motility and chemotaxis and how this compares to overexpression of the properly targeted protein. To this end, we analyzed the phenotype of strains overexpressing the GFP-tagged mistargeted mutants RacH-chim, RacG-chim, RacHΔins and RacH-TM. The last was targeted to the mitochondrial outer membrane by in-frame fusion to the transmembrane domain of Miro (30) and was predominantly membrane associated (Figures 11 B,C, S6). With the exception of RacH-chim, these constructs expressed at levels comparable to RacH-N18 (Figure 1B). Although RacH-chim expressed at low levels, this was sufficient to elicit a phenotype.

RacH-chim cells displayed abundant actin-filled long filopods with occasional branching. In live cells, filopods were frequently very long and highly motile and emanated from lamellipods (Figure 3, phase contrast images). These cells very closely resembled RacG overexpressors (our unpublished results). RacG-chim cells displayed a morphology and pattern of actin distribution similar to AX2 cells. None of the mistargeted constructs showed the strongly impaired growth and pinocytosis defects of RacH-WT (Figures 4B, S2), and only in RacG-chim was phagocytosis severely impaired (Figure 4A). The cytokinesis defect apparent in cells overexpressing RacH-WT growing in suspension was no longer appreciable in cells overexpressing RacG-chim, RacH-chim or RacHΔins (Figure S4). RacH-TM cells, by contrast, were slightly impaired, with 30% cells with three or more nuclei, and this defect was no longer observable in cells grown on a solid substrate. Finally, RacH-chim and RacG-chim cells showed a lower increase in the first peak of actin polymerization upon cAMP stimulation, but the second peak was still appreciable (Figure 10A). However, none of these two strains was severely impaired in its ability to migrate toward the source of chemo-attractant (Figure 10B and Table 1). All this indicates that the phenotype elicited by overexpression of RacH is strongly dependent of its subcellular localization. To further substantiate this hypothesis, we investigated whether re-expression of GFP-tagged mistargeted RacH mutants was able to revert the pinocytosis defect observed in the RacH-KO cells. However, this approach yielded inconsistent results because our attempts resulted in strains that expressed the fusion protein at levels well above those of the endogenous RacH and therefore displayed an overexpressor phenotype (not shown).

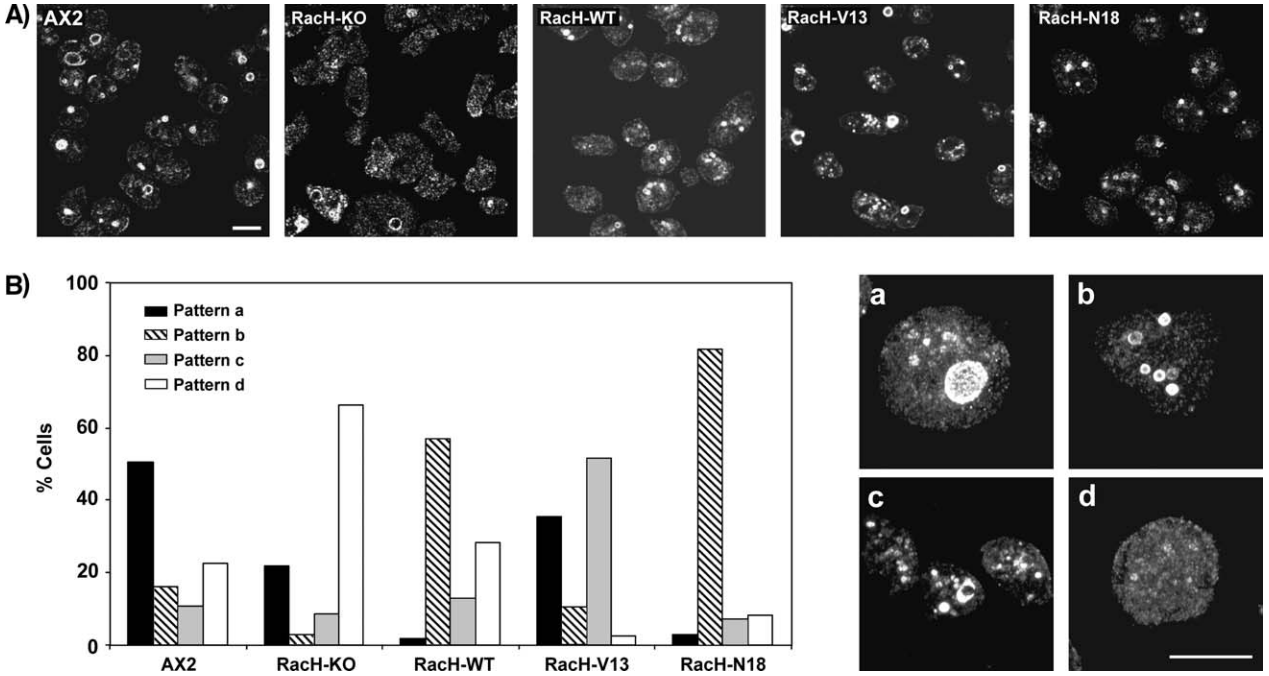

Figure 8: Distribution of vacuolin is disturbed in RacH mutants. A) Cells were allowed to sit on glass coverslips, fixed and labeled with mAb 221-1-1, followed by Cy3-labeled secondary antibody. Images are confocal sections. In RacH-KO cells, the vacuolin-labeled vesicles appear smaller and dispersed over the whole cell. Scale bar, 10 μm. B) Patterns of vacuolin staining in RacH mutants. Cells have been categorized into four classes according to the morphology of their vacuolar system. One example of each class is shown on the right panels: (a) one (occasionally two or three) large prominent vacuole, (b) numerous middle-sized vacuoles, (c) numerous small vesicles, (d) diffuse punctuate pattern. Three hundred cells of each strain have been scored. Scale bar, 10 μm.

Discussion

RacH constitutes the first reported *Dictyostelium* Rho GTPase specifically retained at internal membrane compartments, where it appears to regulate vesicle trafficking processes. The localization of RacH at membranes of the ER and Golgi apparatus along with the defects observed in a RacH-deficient strain are compatible with the proposed role in trafficking events along the endocytic pathway. Although not specifically investigated, indirect evidence suggests that the secretory pathway is unaffected: chemotaxis and development, two processes that depend on maturation and sorting of membrane receptors and adhesion molecules, were not affected in the mutants studied here.

In *Dictyostelium*, the endocytic pathway has been operationally divided into four steps: uptake at the plasma membrane, transit through early acidic endosomes, followed by a late neutral compartment and finally exocytosis (31). We show that RacH-KO cells have defects in all four steps. Although a fraction of RacH appears to localize at the plasma membrane, we attribute the 50% decrease in fluid-phase uptake and the moderate exocytosis defect of the mutant to alterations in trafficking away from the plasma membrane rather than to deficient signaling at this location. This is supported by the fact that targeting of RacH to the plasma membrane (RacH-chim mutant), while stimulating the formation of filopods, does not elicit any alteration in fluid-phase uptake or release. This is in contrast to Rho GTPases that regulate the initial phase of endocytosis by acting at the plasma membrane, like the well-established Rac1, Cdc42 and RhoA in mammalian cells (2,3) and presumably RacF1, Rac1, RacC and RacG in *Dictyostelium* (32–35 and our unpublished observations).

It might seem paradoxical that overexpression of both constitutively active and dominant negative RacH results in a similar phenotype, namely impaired pinocytosis. An inhibitory effect of the same biological function by opposite mutants has already been described. To mention only one example, *Dictyostelium* cells expressing constitutively active or dominant negative Rac1 display inefficient chemotaxis toward cAMP (36). This effect can be interpreted by taking into account how those mutants work in cells. Whereas RacH-N18 might be acting by competing with other Rho GTPases for binding to RhoGEFs, RacH-V13 might be saturating the pool of effectors. Taking into account that overexpression levels are very high, results from RacH-WT, V13 and N18 might not always reflect the physiological role of RacH and must therefore be interpreted very cautiously. Note for example that all three overexpressors bring about a cytokinesis defect, but no such defect is apparent in the knockout strain.

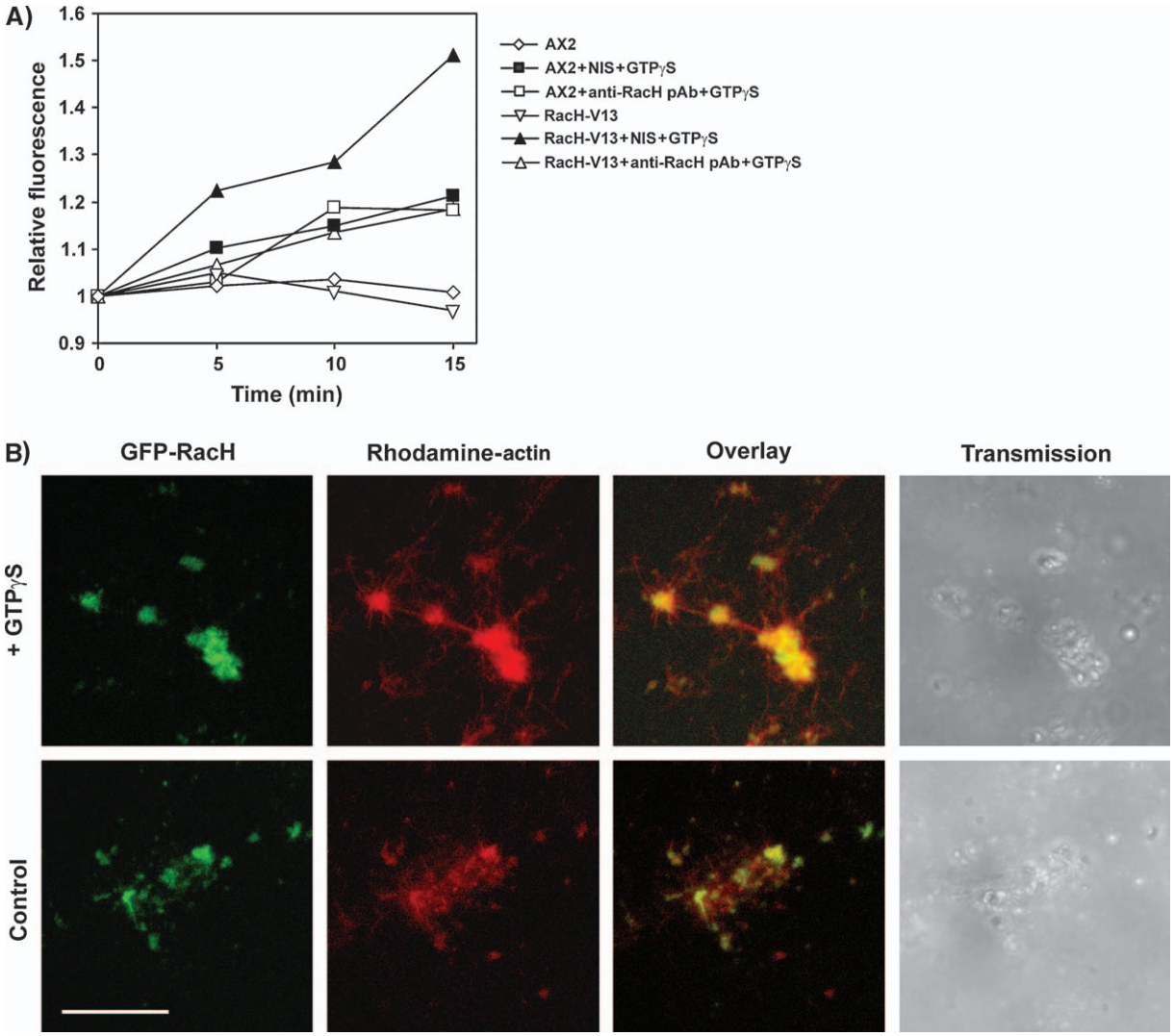

Figure 9: RacH induces actin polymerization. A) Comparison of the actin polymerization response of lysates of AX2 and RacH-V13 cells. The relative F-actin content was determined by TRITC-phalloidin staining of samples fixed at the indicated time points after induction with 100 μM GTPγS. The amount of F-actin was normalized relative to the F-actin level of the corresponding uninduced lysate. Where indicated, samples were preincubated for 30 min with RacH-specific polyclonal antiserum or with nonimmune serum (NIS) at a 1:100 dilution prior to GTPγS addition. All values are the average of at least three independent experiments, each performed in triplicate. For the sake of clarity, error bars are not shown. The coefficient of variation of these experiments was 10%. B) Visual assay. A lysate of cells expressing GFP-RacH-V13 was prepared. To this, rhodamine-labeled actin, creatine phosphate and ATP were added. Upon addition of GTPγS actin, polymerization was stimulated. Images were taken with a confocal laser scanning microscope. Bar, 10 μm.

RacH-KO cells also showed deficient acidification of early endosomes, a process driven by accumulation of vacuolar H^+-ATPase. Clarke et al. (37) have shown that in *Dictyostelium*, most of the proton pumps delivered to a new endosome are retrieved from late endosomes. Concentration of fluid-phase marker in postlysosomal compartments is not altered in RacH-KO, suggesting that retrograde transport of the pump is not affected. Taken together, our data pinpoint the role, or one of the roles, of racH to the delivery of vesicles carrying proton pumps to early endosomes. If RacH is involved specifically in this step, it is not surprising that neither the protein levels nor the distribution of VatA, a component of the vacuolar proton pump, was noticeably altered in RacH-KO because most of the proton pump is localized on vesicles of the contractile vacuole system, a compartment that apparently does not exchange membranes with the endolysosomal compartment (37). The lower amounts of secreted acid phosphatase observed in RacH-KO cells in the presence of sucrose might be related to a deficient acidification of endosomes: previous reports on *Acanthamoeba* indicated that hydrolase transport and secretion is pH dependent (38).

RacH is necessary for proper distribution of vacuolin, a marker for the postlysosomal compartment of *Dictyostelium*. Vacuolin, which exists as two isoforms, is a

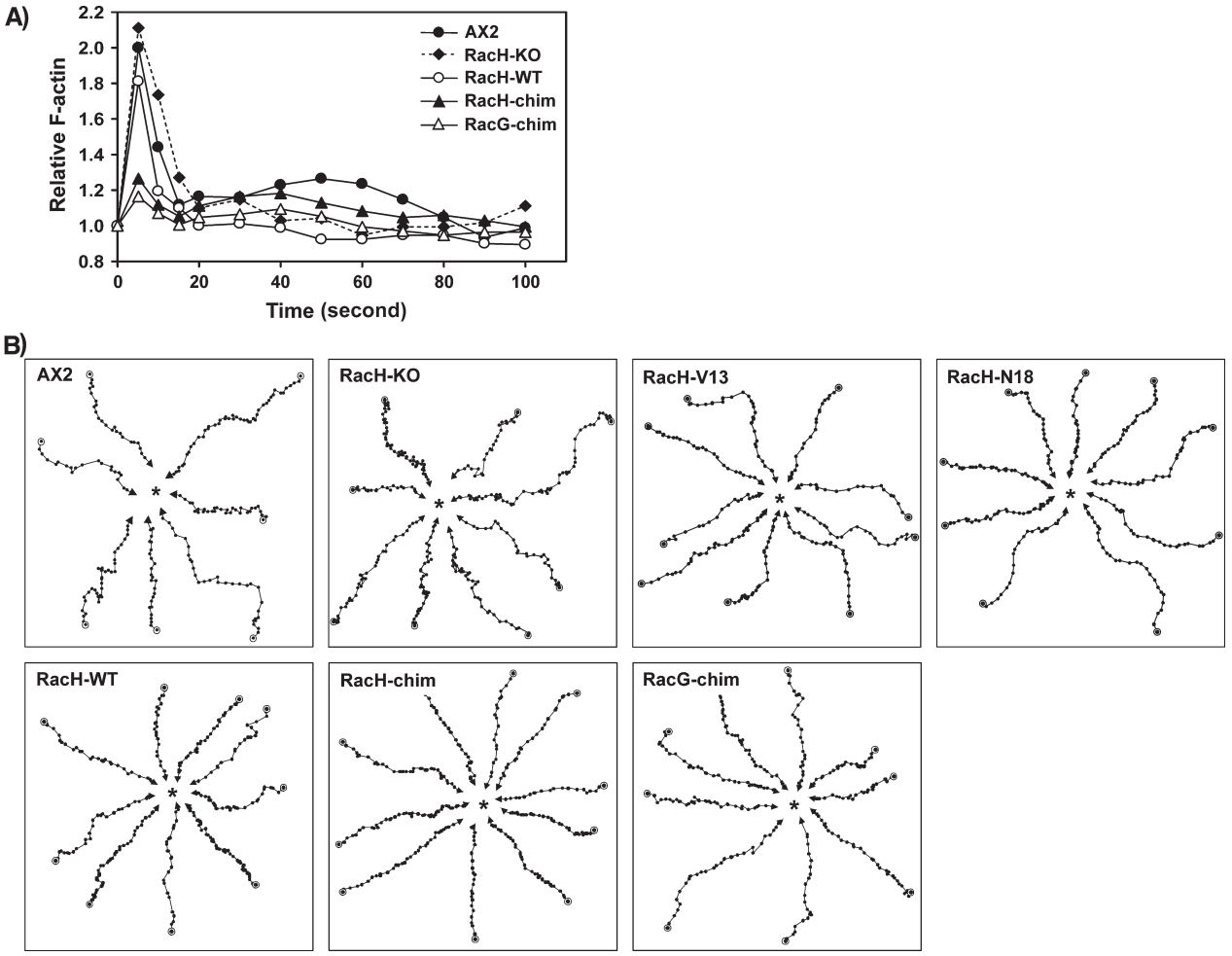

Figure 10: Chemotaxis of aggregation competent RacH mutant cells. A) Actin polymerization response upon cAMP stimulation. The relative F-actin content was determined by TRITC-phalloidin staining of cells fixed at the indicated time points after stimulation with 1 μM cAMP. The amount of F-actin was normalized relative to the F-actin level of unstimulated cells. B) Chemotactic movement to a micropipette containing cAMP. Cells were starved for 6 h, allowed to sit on a glass coverslip and stimulated with a micropipette filled with 0.1 mM cAMP. Images of chemotaxing cells were captured every 30 seconds. Cell movement was analyzed with the DIAS software.

member of the PHB (prohibitin homology) domain family that also includes flotillins and related proteins of higher eukaryotes frequently associated with lipid rafts (39). While vacuolin A is dispensable, disruption of vacuolin B results in markedly impaired exocytosis, enlargement of the neutral postlysosomal compartment and accumulation of the fluid-phase marker (40). Except for a moderate exocytosis defect, none of these defects was observed in RacH-KO cells, suggesting that in spite of its altered distribution, function of vacuolin is not affected. The predominantly diffuse pattern of vacuolin distribution seen in RacH-KO suggests that RacH is necessary for accumulation of vacuolin at larger vesicles prior to exocytosis. Targeting of vacuolin has not been studied in detail, but experiments using latrunculin A or a hybrid protein that targets cofilin to the postlysosomal compartment have revealed a role of the F-actin cytoskeleton in this process (41). It is interesting to note that other proteins of the PHB domain family like stomatin and podocin also colocalize partially with the actin cytoskeleton (39).

The mechanism of RacH action is not clear at the moment. Like Cdc42, RacH could be involved in vesicle transport within and from the Golgi apparatus by promoting actin-driven vesicle movement (42). RacH is able to induce actin polymerization, and it is well established that progression of vesicles along the endocytic pathway depends on actin (40,41). However, RacH does not interact with WASP or any other effector carrying a CRIB domain (26,43 and our unpublished observations). RacH does also not interact with IQGAP-related proteins (our unpublished observations), which have been reported to associate with Cdc42 at Golgi membranes (44). In mammalian cells, RhoD regulates motility of early endosomes through interaction with a Diaphanous-related formin (4); therefore, formins may constitute a candidate family of RacH effectors to be addressed in future studies.

Table 1: Analysis of cell motility of RacH mutants[a]

	AX2	RacH-KO	RacH-WT	RacH-V13	RacH-N18	RacH-chim	RacG-chim
Buffer							
Speed (μm/min)	5.61 ± 3.37	5.98 ± 2.57	4.38 ± 1.96	4.74 ± 2.21	6.44 ± 4.26	5.30 ± 2.52	5.47 ± 2.37
Persistence (μm/min × degree)	1.71 ± 1.59	1.55 ± 1.07	1.02 ± 0.59	1.71 ± 1.26	2.22 ± 1.86	1.78 ± 1.46	1.59 ± 0.93
Directionality	0.40 ± 0.25	0.32 ± 0.21	0.29 ± 0.17	0.53 ± 0.29	0.43 ± 0.25	0.45 ± 0.26	0.56 ± 0.23
Directional change (degree)	47.07 ± 18.71	57.92 ± 17.06[b]	53.11 ± 11.03	40.78 ± 23.27	53.62 ± 17.89	44.89 ± 17.21	35.43 ± 16.34
cAMP							
Speed (μm/min)	11.83 ± 3.30	8.94 ± 2.83[b]	8.04 ± 2.51[b]	11.55 ± 2.78	9.57 ± 2.37[b]	9.60 ± 2.71[b]	9.40 ± 2.84[b]
Persistence (μm/min × degree)	3.96 ± 2.04	2.65 ± 1.39[b]	2.72 ± 1.26[b]	3.67 ± 1.17	3.26 ± 1.30	3.28 ± 1.26	3.23 ± 1.28
Directionality	0.72 ± 0.19	0.52 ± 0.23[b]	0.76 ± 0.16	0.72 ± 0.18	0.81 ± 0.09	0.83 ± 0.14	0.72 ± 0.21
Directional change (degree)	24.61 ± 10.77	43.21 ± 18.08[b]	25.46 ± 13.02	22.50 ± 12.50	20.55 ± 8.22	19.27 ± 10.84	25.31 ± 13.07

[a]Time-lapse image series were captured and stored on a computer hard drive at 30-second intervals. The DIAS software was used to trace individual cells along image series and calculate motility parameters. Objects whose speed was <2 μm/min were excluded from the analysis. Persistence is an estimation of movement in the direction of the path. Directionality is calculated as the net path length divided by the total path length and gives 1.0 for a straight path. Directional change represents the average change of angle between frames in the direction of movement. Values are mean ± standard deviation of 30–90 cells from at least three independent experiments.
[b]$p < 0.05$ relative to AX2 in the same condition (analysis of variance). For every strain, all parameters differed significantly between the buffer and the cAMP condition.

In this study, we also analyzed the importance of the CAAX motif and the second signal in the hypervariable region for targeting and functionality of *Dictyostelium* RacH. RacH is predominantly cytosolic, but 40% of the protein is targeted to the perinuclear region, Golgi apparatus and ER, closely resembling the behavior of human Cdc42 and Rac2 (15). We initially attributed this localization of RacH to lack of a functional second signal: negatively charged residues in this region lower the net positive charge to +2 (Figure 11A). It has been shown that a net positive charge of 4 or more is required to target K-ras4B to the plasma membrane (14,45), and indeed, human Cdc42 and Rac2, which have a weak polybasic region, are targeted predominantly to internal membranes (15). By contrast, in *Dictyostelium* Rho GTPases that are targeted to the plasma membrane, like Rac1a/1b/1c (33), RacC (46), RacE (47), RacF1 (32) and RacG, the hypervariable region has charges that range from +4 to +8, similar to that in human Rac1 (15). Chimeric constructs of RacG and RacH in which the C-terminal region was exchanged appeared to confirm the role of the net positive charge of the hypervariable region: the RacH chimeric construct was targeted to the plasma membrane, whereas the RacG chimeric construct was retained at endomembranes. This is in-line with the results obtained with human Cdc42 and TC10 (a Cdc42-related GTPase). TC10 possesses both a palmitoylation signal and a polybasic stretch and localizes on the plasma membrane and endosomes. A chimeric construct in which the C-terminal 20 amino acids were replaced by the homologous region of Cdc42 displayed a localization identical to Cdc42 (15). These and other results reported by Michaelson et al. (15) led to propose that subcellular localization of Rho GTPases is determined primarily by the hypervariable region and the capacity to bind RhoGDI. A role for RhoGDI in the localization of RacG or RacH, however, can be excluded because neither of both proteins interacts with RhoGDI (48). A more detailed analysis using alanine-exchange mutants revealed that further sequences upstream of the polybasic region within the 38 residues exchanged in the chimeric mutants contribute to targeting of RacH: the RacH-4Ala and the RacG-chim-4Ala mutants, although carrying a net positive charge of +6, were still substantially retained at endomembranes.

One more determinant of the subcellular localization of RacH appears to be the insert region. Although this region does not change conformation upon activation, in combination with the effector domain, it contributes to determine the specificity of interactions of Rho GTPases (49). For example, although not required for binding to, it is required for activation of the nicotinamide adenine dinucleotide phosphate, reduced form, (NADPH) oxidase complex by Rac (50) or phospholipase D1 by Cdc42 (51). Altogether, our results indicate that regions other than the hypervariable region are responsible for proper targeting of Rho GTPases, and this might be a more general principle applicable to lipid-modified GTPases of the Ras superfamily, as has been shown recently for Rab GTPases in an extensive study (52).

The fact that RacG and RacH display radically different subcellular localizations and play different roles was exploited to investigate to which extent targeting of the GTPase to a particular region or compartment determines its biological functions. RacG regulates mainly processes that require actin remodeling at the cell cortex: overexpression of RacG elicits extension of filopods, stimulates particle uptake and impairs chemotaxis (our unpublished observations). Targeting of RacH to the

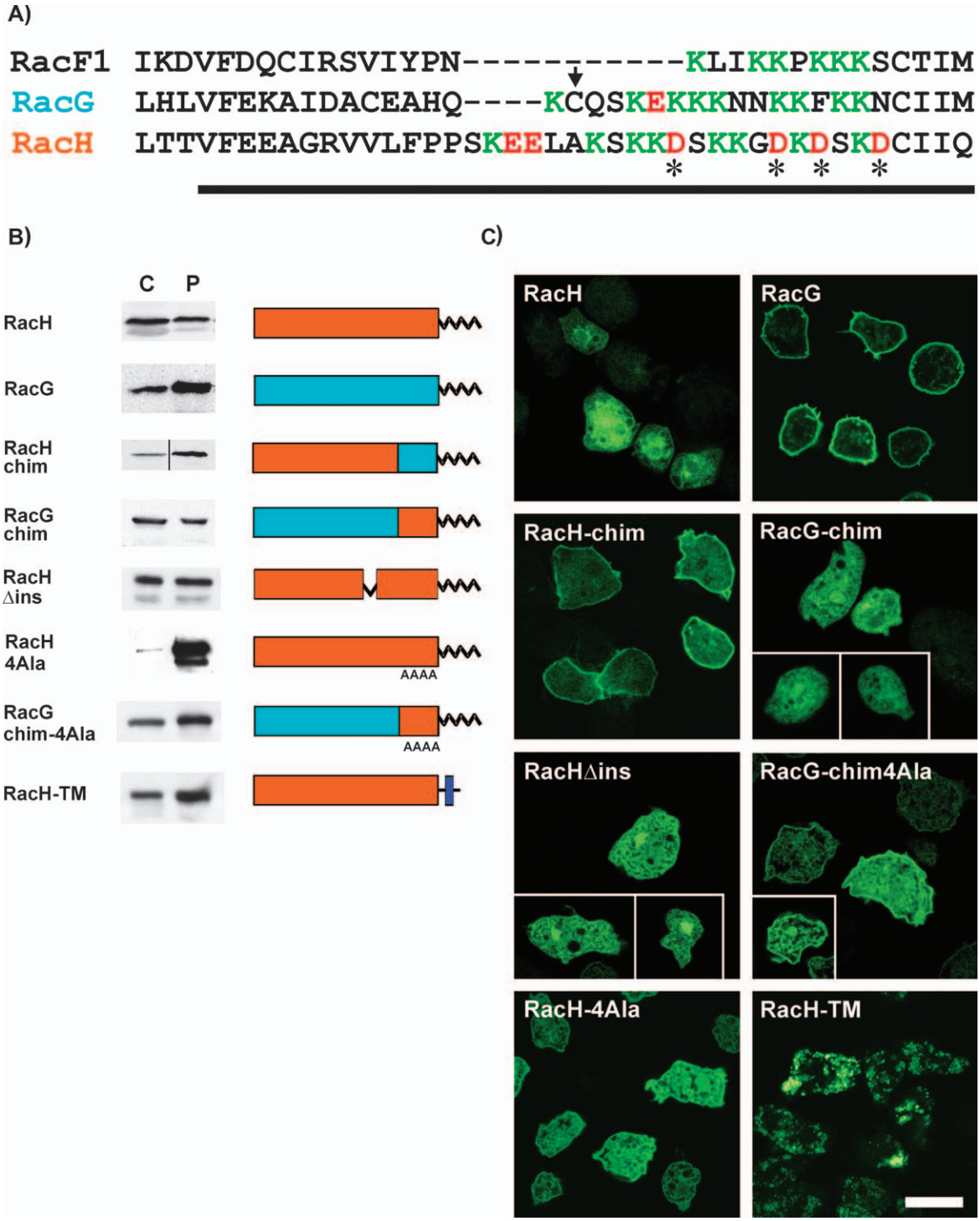

Figure 11: Determinants of RacH targeting. A) The C-terminal membrane association region of RacH in comparison with RacF1 and RacG. All three Rho proteins end with a CAAX prenylation motif. Note the presence of numerous negatively charged residues (red) in the polybasic stretch (green) of RacH that have been neutralized by alanine exchange (asterisks). The cysteine 183 of RacG that has been mutated to serine is indicated with an arrow. The bar below the sequences denotes the part exchanged in the chimeric constructs. B) Fractionation of *Dictyostelium* cells overexpressing GFP fusions of RacH, RacG, and diverse constructs as indicated on the right. Cells were lysed by sonication, and cytosolic (C) and particulate (P) fractions were separated by ultracentrifugation. Samples were resolved in 12% polyacrylamide gels and blotted onto nitrocellulose membranes. Blots were incubated with anti-GFP mAb K3-184-2. C) Confocal sections of *Dictyostelium* cells expressing the GFP fusions indicated in B. Cells were fixed with cold methanol. Bar, 10 μm.

plasma membrane was accompanied by extension of filopods and impaired F-actin response to cAMP but fairly normal chemotactic response. By contrast, targeting of RacG to internal membranes impaired phagocytosis but had no effect on the rate of growth and pinocytosis or on cytokinesis, which are severely affected in RacH-WT cells. This indicates that RacH is able to interact with effectors and/or regulatory elements of other Rho GTPases at the plasma membrane but is prevented of doing so by retention at inner membrane compartments. The conclusion that targeting to the appropriate membrane compartment is necessary for the Rho GTPase to elicit its functions was further supported by results from RacHΔins and RacH-TM mutants: overexpression of these mutated RacH proteins did not recapitulate any of the phenotypes elicited by overexpression of RacH-WT (the moderate cytokinesis defect of RacH-TM might result from alterations of the mitochondrial function). Reports that address the importance of membrane targeting of Rab GTPases also underscore our conclusion. Rab27a, a protein that associates specifically with melanosomes, did not rescue the coat color phenotype of a Rab27a null mouse when mistargeted to the ER (53), and in yeast, mistargeted lipid tail variants of two essential Rab proteins, Ypt1p and Sec4p, were not able to complement thermosensitive alleles of the respective gene (54). The fact that RacH-TM does not induce accumulation of actin around mitochondria (Figure S6) favor the explanation that the effects elicited by RacH depend on the interaction with regulators and effectors restricted to specific locations within the cell.

In summary, we show that RacH is involved in regulation of vesicle trafficking, more specifically in sorting of vesicles from the secretory pathway to several compartments along the endocytic pathway, presumably by virtue of its effects on actin polymerization. We also show that several regions of the molecule, not only the hypervariable region, determine targeting of RacH and that targeting to the correct membrane compartment probably allows interactions with appropriate regulators and effectors and is therefore essential for function.

Materials and Methods

Strains, growth conditions, and development of D. discoideum

Cells of *D. discoideum* strain AX2 (referred to as WT) and transformants were grown either in liquid nutrient medium at 21°C, with shaking at 160 r.p.m. (55), or on nutrient agar plates with *Klebsiella aerogenes* (56). For development, the cells were grown to a density of 2×10^6 to 3×10^6 cells/mL, washed in 17-mM Soerensen phosphate buffer of pH 6.0, and 0.5×10^8 cells were deposited on phosphate agar plates and allowed to develop by incubating at 21°C as described (57).

Generation of a racH *knockout strain*

To construct the *racH* disruption vector, DNA fragments containing the 5′ untranslated region (UTR) of *racH* were amplified from genomic DNA using primers PracH: −837 (5′-GTCGACATAGTAATAGCAACAATCATAATGG-3′), and PracH: 27 (5′-AAGCTTAACCATTACTTTAATATCTTTTACC-3′), and digested with *Sal*I and *Hin*dIII. DNA fragments containing the 3′UTR region of *racH* were amplified using primers PracH: 1885 (5′-TCTAGAAAAAAAGGT-GATAAAGACTCAAAGG-3′), and PracH: +910 (5′-GCGGCCGCACCAAG-TAATGATAGTATAGAATCC-3′), and digested with *Xba*I and *Not*I. Blasticidin-S-resistance (Bsr) cassette (58) was excised from pLPBLPv2 (M. Landree, Johns Hopkins University, Baltimore, unpublished data) using *Hin*dIII and *Xba*I. These three DNA fragments were cloned into pBluescript (Stratagene, La Jolla, CA, USA), forming the disruption vector pRacHdis (Figure S1A). After digestion with *Sal*I and *Not*I, linearized pRacHdis (10 μg) was introduced into growth phase of AX2 cells by electroporation. Transformants were selected in nutrient medium containing 5 μg/mL blasticidin (ICN Biomedicals Inc., Aurora, OH, USA) on plastic dishes. After 5 days, the cells were harvested and cloned on SM agar plates with *K. aerogenes*. Resultant colonies were checked for disruption of the *racH* gene by polymerase chain reaction (PCR) and Southern blot analysis (Figure S1B).

Mutagenesis of RacH

Dictyostelium RacH sequences carrying M13V (constitutively active) or T18N (dominant negative) mutations were generated from WT complementary DNA (cDNA) by PCR-based, site-directed mutagenesis (26,43). PCR-based mutagenesis with mismatched reverse primers was used to generate a Q200M mutant of RacH, a C183S mutant of RacG, and mutants in which aspartic acid residues were substituted by alanine residues (Figure 11A).

Chimeric PCR products were generated by two PCR steps. In the first step, N-terminal fragments of RacG (amino acids 1–168) and RacH (amino acids 1–163) and C-terminal fragments of RacG (amino acids 167–201) and RacH (amino acids 164–200) were amplified (Figure 11A). The oligonucleotide primers were designed to allow an overlap of 15 bases between the N-terminal fragment of one GTPase and the C-terminal fragment of the other. Finally, the full-length chimeric cDNAs were amplified using a RacG forward primer and a RacH reverse primer to obtain RacG chimera and a RacH forward primer and a RacG reverse primer to obtain RacH chimera. To generate a RacH lacking the Rho-insert region (residues 125-131; RacHΔins), overlaping fragments upstream and downstream of the insert were amplified separately by PCR on a RacH template and joined in a second PCR step. A mitochondrial-membrane-targeted RacH was generated by fusing in-frame RacH to the C-terminal transmembrane region (amino acids 603–658) of *Dictyostelium* Miro using conventional cloning of PCR-amplified fragments.

Polymerase chain reaction products were cloned into the pGEM-Teasy vector (Promega, Madison, WI, USA) and verified by sequencing, done at the service laboratory of the Center for Molecular Medicine, Cologne, using an automated sequencer (ABI 377 PRISM; Perkin Elmer, Norwalk, CO, USA).

Vectors were constructed that allowed expression of the red-shifted S65T mutant of *Aequoria victoria* GFP fused to the amino terminus of RacH in *Dictyostelium* cells. Because the carboxyl terminus of Rho-related proteins contains structural elements responsible for membrane association (45), a fusion of GFP at the amino-terminal end of RacH was chosen. For overexpression of mutated variants V13 and N18, a tetracycline-controlled inducible system was used (59). Fusions to GFP were prepared in pBluescript (Stratagene) and blunt end cloned into the MB38 plasmid. This vector was introduced into AX2 cells carrying the MB35 vector. For overexpression of the WT form of RacH and all other mutants, cDNA fragments were cloned in the transformation vector pDEX-GFP under the control of the actin-15 promoter (60). This vector confers G418 resistance and was introduced into AX2 cells. All vectors were introduced into cells by electroporation. G418 (Sigma, Deisenhofen, Germany) and blasticidin were used for selection. GFP-expressing transformants were confirmed by visual inspection under a fluorescence microscope.

Antibody generation

A PCR-amplified cDNA fragment encoding RacH was cloned into the *Bam*HI site of the bacterial protein expression vector pQE30 (Qiagen GmbH, Hilden, Germany). This vector was transformed into *Escherichia*

coli strain M15(pREP4). The recombinantly expressed His-tagged protein was purified using affinity chromatography with Ni^{2+}-NTA agarose beads, and 100 μg of protein was used to immunize two female, white New Zealand rabbits (Pineda Antikörper-Service, Berlin, Germany), followed by two boosts of 100 μg each at 2-week intervals. Specificity of the antiserum was verified against a panel of glutathione S-transferase fusions of *Dictyostelium* Rho GTPases.

Western blotting

The SDS-PAGE and Western blotting were performed using standard methods. An enhanced chemiluminescence detection system (Amersham Pharmacia Biotech, Freiburg, Germany) was used. GFP was detected with specific mAb K3-184-2, actin with mAb Act 1-7 (61), and α-actinin with mAb 47-62-17 (62).

Cell fractionation experiments

GFP-RacH cells were collected by centrifugation (1000× ***g*** for 5 min) and resuspended at a density of 2×10^8 cells/mL in MES buffer (20 mM 2-[N-morpholino] ethane sulfonic acid, 1 mM ethylenediaminetetraacetic acid, 250 mM sucrose, pH 6.5) supplemented with a protease inhibitor mixture (Roche Diagnostics, Mannheim, Germany). Cells were homogenized on ice by sonication and light microscopy was performed to ensure that at least 95% of the cells were broken. Cytosolic and particulate fractions were separated by ultracentrifugation (100 000×***g*** for 30 min).

Conventional and fluorescence microscopy

To record distribution of GFP fusion proteins in living cells, cells were grown to a density of 2×10^6 to 3×10^6 cells/mL and transferred onto 18-mm glass coverslips with a plastic ring for observation. For studies on fixed cells, cells were fixed either in cold methanol (−20°C) or at room temperature with picric acid/paraformaldehyde (15% v/v of a saturated aqueous solution of picric acid/2% paraformaldehyde, pH 6.0), followed by 70% ethanol. Actin was detected using mAb Act 1-7 (61), interaptin using mAb 260-60-10 (63), PDI using mAb 221-135-1 (64), comitin using mAb 190-340-2 (65), the A subunit of the V/H^+-ATPase using mAb 221-35-2 (40), a common mannose-6-sulfate-containing carbohydrate epitope present in lysosomal enzymes using mAb 221-342-5 (66), and vacuolin using mAb 221-1-1 (67), followed by incubation with Cy3-labeled anti-mouse immunoglobulin G. Nuclei were stained with DAPI (Sigma-Aldrich, St. Louis, MO, USA). Mitochondria were stained with MitoTracker Red CMXRos (Molecular Probes Europe, Leiden, Netherlands) for 45 min prior to fixation.

Confocal images were taken with an inverted Leica TCS-SP laser scanning microscope with a 100× HCX PL APO NA 1.40 oil immersion objective (Leica Microsystems, Wetzlar, Germany). For excitation, the 488-nm argon-ion laser line and the 543-nm NeNe laser line were used. Images were processed using the accompanying software.

Actin polymerization assay

Chemo-attractant-induced F-actin formation in aggregation competent cells was quantitated as described (25). Briefly, cells were resuspended at 2×10^7 cells/mL in Soerensen buffer and starved for 6–8 h. Cells were stimulated with 1 μM cAMP and 50 μL of samples were taken at various time points. The reaction was terminated by addition of 450 μL stop solution (3.7% formaldehyde, 0.1% Triton-X-100 (Sigma-Aldrich, St. Louis, MO, USA), 0.25 μM TRITC-phalloidin in 20 mM potassium phosphate, 10 mM Pipes, 5 mM EGTA, 2 mM $MgCl_2$, pH 6.8). After staining for 1 h, samples were centrifuged for 5 min at 15 000×***g***. Pellets were extracted using 1 mL methanol for 16 h, and fluorescence (540/565 nm) was read in a PTI fluorimeter (Photon Technology Intl., Seefeld, Germany).

For quantification of actin polymerization in a cell-free system, lysates were prepared as described (24). Briefly, cells were starved as described above, resuspended at 3×10^8 cells/mL in cold PM (10 mM phosphate buffer of pH 6.1, 2 mM $MgSO_4$, 20 mM KCl, 1 mM EGTA) and lysed on ice by sonication. Actin polymerization was induced by addition of 100 μM GTPγS. F-actin was quantitated by TRITC-phalloidin staining of formaldehyde-stabilized pelleted material as indicated above, with the difference that 0.4 μM TRITC-phalloidin was contained in the stop solution.

For visualization of actin polymerization, a modification of the method of Ma et al. (68) was used. Cells were resuspended at 1.5×10^8 cells/mL in lysis buffer (10 mM Hepes, pH 7.7, 100 mM KCl, 2 mM $MgCl_2$, 0.1 mM $CaCl_2$, 5 mM EGTA, 50 mM sucrose, 1 mM DTT and protease inhibitor mixture) and lysed by sonication. About 7.5 μL of cell extract was mixed with 1 μL of rhodamine-labeled rabbit muscle actin (0.3 mg/mL; Cytoskeleton, Denver, CO, USA) and 0.5 μL of ATP regeneration mix (150 mM creatine phosphate, 20 mM ATP, 2 mM EGTA, 25 mM $MgCl_2$). After addition of 1 μL GTPγS (final concentration of 1 mM), 5 μL of the reaction mixture was deposited on a coverslip and squashed with a slide for microscopy.

Chemotaxis assay

Vegetative cells were resuspended at 1×10^7 cells/mL in Soerensen phosphate buffer and starved for 6–8 h. Cells were then transferred onto a 5-cm glass coverslip with a plastic ring placed on an Olympus IX70 inverse microscope equipped with a 10× UplanFl 0.3 objective (Olympus, Hamburg, Germany). Cells were stimulated with a glass capillary micropipette (Eppendorf Femtotip) filled with 0.1 M cAMP (69). Time-lapse image series were captured and stored on a computer hard drive at 30-second intervals with a JAI CV-M10 CCD camera and a Imagenation PX610 frame grabber (Imagenation Corp., Beaverton, OR, USA) controlled through Optimas software (Optimas Corp., Bothell, Washington). The DIAS software (Soltech, Oakdale, IA, USA) was used to trace individual cells along image series and calculate the cell motility parameters (70).

Endocytosis

Phagocytosis was assayed as described using TRITC-labeled yeast cells (71). Fluid-phase endocytosis and exocytosis assays were performed using TRITC-dextran or FITC-dextran according to Aubry et al. (72).

Secretion of lysosomal enzymes

Cells were harvested by centrifugation and resuspended at a concentration of 1×10^7 cells/mL in Soerensen buffer with or without addition of sucrose (100 mM). One-milliliter samples was harvested by centrifugation at the times indicated. Enzyme assays were performed on the supernatant (extracellular) and on cells lysed in the presence of 0.1% Triton-X-100. The acid phosphatase and α-mannosidase activities were measured as described (22).

Determination of endolysosomal pH

The endosomal pH was measured by a dual excitation ratio method (excitation at 450 and 495 nm and emission at 520 nm) using FITC-dextran (70 000 Mr; Sigma-Aldrich, St. Louis, MO, USA) (2 mg/mL) or FITC-labeled *E. coli* B/r as pH probes as described by Temesvari et al. (21). Briefly, the cells were grown to a density of 2×10^5 to 5×10^5 cells/mL, harvested and resuspended at a concentration of 3×10^6 cells/mL in fresh axenic medium, and loaded with the pH probe. For measurement, the cells were collected by centrifugation, washed and resuspended in 50 mM MES buffer, pH 6.5, and the fluorescence intensity was measured using a PTI fluorimeter. To monitor changes of endosomal pH during transit of the fluid or solid-phase markers, the cells were pulsed with either FITC-dextran or FITC-labeled bacteria for 10 min, washed and resuspended in fresh nutrient medium. The pH was determined from an *in vitro* standard curve of FITC-dextran in a pH range of 4–7.

To monitor changes of pH and concentration of marker during endocytosis (40), vegetative cells were washed with Soerensen buffer and allowed to sit in an observation chamber made of a glass coverslip with a plastic ring. Cells were then exposed to pH-sensitive FITC- and pH-independent TRITC-dextran for 1 h, and single confocal sections were taken as indicated above.

Miscellaneous methods

Standard molecular biology methods were applied as described by Sambrook and Russell (73).

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Supplemental Materials

Figure S1: Disruption of the *racH* gene. A) Diagram of the *Dictyostelium racH* gene (blue) and its disruption by the bsr. A targeting vector was used, with the blasticidin cassette flanked on the left by a 850-bp- long arm homologous to the 5′ region and on the right by a 960-bp long arm homologous to the 3′ region of *racH*. Red arrows represent genes upstream and downstream of *racH*. B) Screening for homologous recombinants was performed by Southern blotting. Genomic DNA was digested with *Eco*RI and the blot probed with a ^{32}P-labeled probe in the 5′ region of the *racH* gene. A shift of the 2.4-kb WT band to a 3-kb band in the knockout is apparent.

Figure S2: Growth of RacH mutants in suspension. Cultures were inoculated at 2×10^5 cells/mL and grown at 21°C, with shaking at 160 r.p.m. Cells were counted at the indicated time points. The RacH-WT overexpressor shows impaired growth. Data are the average of three independent experiments.

Figure S3: Distribution of membrane compartments in cells lacking RacH. AX2 and RacH-KO cells were fixed in cold methanol and incubated with mAbs against the following membrane compartment markers: the Golgi marker comitin (mAb 110-340-2), the ER marker PDI (mAb 221-135-1), a marker for the contractile vacuole/endolysosomal system (vatA, mAb 221-35-2), and a marker for the lysosomal compartment (a carbohydrate antigen, mAb 221-342-5). Images are confocal sections. Bar, 10 μm.

Figure S4: Cytokinesis of RacH mutants. A) Fluorescence and phase contrast images of cells grown in shaking culture. Cells were allowed to sit for 20 min on coverslips and fixed with picric acid/paraformaldehyde, and nuclei were stained with the DNA-binding dye DAPI. Arrows indicate multinucleate cells. RacH-V13 and RacH-N18 cells (not shown) resembled RacH-WT. Scale bar 25, μm. B) Quantification of nuclei of cells grown in shaking suspension. Samples were processed as in A. For every strain, nuclei of 250–300 cells were counted.

Figure S5: Development of cells lacking RacH. Cells were starved at a cell density of 5×10^7 on phosphate agar plates, and images were taken at the indicated time points using a stereomicroscope. RacH-KO cells show the same developmental behavior as AX2 cells. Bar, 1 mm.

Figure S6: Mitochondrial-membrane-targeted RacH. GFP-RacH was targeted to the outer mitochondrial membrane by a fusion to the C-terminal transmembrane region of Miro. Cells were fixed with paraformaldehyde. In order to visualize the mitochondria, cells were incubated with MitoTracker Red prior to fixation. Actin was visualized with TRITC-phalloidin. Pictures are confocal sections. Bar, 10 μm.

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